

15936 – Overly permissive frame navigation allows password theft

15936 – Overly permissive frame navigation allows password theft - Adam Barth, WebKit Bugzilla.

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Content

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15936 – Overly permissive frame navigation allows password theft

Field	Value
Status	RESOLVED FIXED
Product / component	WebKit / Frames
Version	528+ (Nightly build)
Priority / severity	P2 / Normal
Hardware / OS	All / All
Reported	2007-11-10 16:45 PST
Last modified	2008-01-03 22:28 PST
Assignee	Nobody
Keywords	InRadar
URL	AdSense demonstration

Attachments

Attachment	Date	Author	Review status
Updates the frame navigation policy to match Internet Explorer 7 (single window) (2.64 KB, patch)	2007-11-28 17:26 PST	Collin Jackson	Sam Weinig: review-

[Details](#) · [Formatted Diff](#) · [Diff](#)

Report

Adam Barth — 2007-11-10 16:45:31 PST

WebKit's frame navigation policy is overly permissive, allowing a web attacker to steal passwords from many sites including Google and several banks.

WebKit allows any page to navigate subframes of any other page. If a site embeds a password field in an frame, a malicious web site operator can navigate that frame to his site and steal user passwords.

The steps below can be used to reproduce the issue:

1. Navigate to <http://crypto.stanford.edu/~abarth/research/webkit/adsense.html>
2. Click "Open AdSense in a new window."
3. Click "Navigate the AdSense login frame."
4. Notice the password field in the AdSense window has been replaced by content of the attackers choice. The address bar reads "www.google.com" and the lock icon is intact.

The frame navigation policy in Firefox 2 was developed in 1999 in response to a similar attack against CitiBank [1]. Their policy is as follows:

- Allow the navigation if the source and target frames contained in the same window.
- Allow if the source frame can script the target frame or one of its ancestors in the frame hierarchy.

Internet Explorer 7 is more strict than Firefox 2. For example, IE7 forbids the navigation from the lower frame in [2] whereas Firefox 2 permits it. From what we can tell, IE7 is enforcing the following policy:

- Allow if the source frame can script the target frame or one of its

ancestors in the frame hierarchy.

The HTML5 spec [3] is the most strict. From our reading, it forbids both of the navigations in [4], whereas all the browsers we've tested allow both.

We recommend WebKit implement the same frame navigation policy as Internet Explorer 7:

- Allow if the source frame can script the target frame or one of its

ancestors in the frame hierarchy.

References:

[1] https://bugzilla.mozilla.org/show_bug.cgi?id=13871 [2] <http://crypto.stanford.edu/~abarth/research/nav/frame1.html> [3] <http://www.whatwg.org/specs/web-apps/current-work/#the-rules> [4] [http://www.whatwg.org/specs/web-apps/current-work/#the-rules](#)

Comment 1

Mark Rowe (bdash) — 2007-11-10 17:24:39 PST

<rdar://problem/5592988>

Comment 2

Sam Weinig — 2007-11-26 16:16:29 PST

Fixed in [r28056](#).

Comment 3

Adam Barth — 2007-11-27 00:11:59 PST

Thanks for addressing this quickly. According to the comments in your patch, you implemented the following frame navigation policy:

The navigation change is safe if the active frame is:

- in the same security domain (satisfies same-origin policy)
- the opener frame
- an ancestor or a descendant in frame tree hierarchy

I'm curious why you decided on this policy. It doesn't match Firefox or Internet Explorer 7. For example, the "Navigate middle frame" link at <<http://xenon.stanford.edu/~abarth/research/nav/frame1.html>> is forbidden by IE7 but allowed by this policy.

If we can reach a consensus frame navigation policy among the major browsers, then we can standardize the behavior as part of HTML5 and the web at large will benefit.

Comment 4

Sam Weinig — 2007-11-27 09:31:21 PST

I allowed navigating the ancestor frame because when I tested IE I noticed it allowed the parent frame (in another domain) to be navigated. It seems my testing and analysis was too limited and the behavior of IE is actually that only the frame at the top of the hierarchy is allowed to be navigated. Does this match your findings?

Comment 5

Adam Barth — 2007-11-27 13:22:18 PST

the behavior of IE is actually that only the frame at the top of the hierarchy is allowed to be navigated. Does this match your findings?

Yes, that's my current understanding as well. I'll do some more systematic testing of IE7 and report back by the end of the week.

Comment 6

Collin Jackson — 2007-11-28 17:26:43 PST

Created [attachment 17584](https://bugs.webkit.org/attachment.cgi?id=17584&action=edit) [[details]](<https://bugs.webkit.org/attachment.cgi?id=17584&action=edit>) Updates the frame navigation policy to match Internet Explorer 7 (single window)

We did some more exhaustive testing of frame navigation in Internet Explorer 7 and wrote a patch to update WebKit to match IE7.

Here is a test framework that we created: <http://w3sim.com/frames/> Here are the results on IE7: <http://w3sim.com/frames/screenshots/ie7-single.png> Here are the results on the nightly WebKit: <http://w3sim.com/frames/screenshots/webkit-2007-11-27-single.png> Here are the results after the attached patch: <http://w3sim.com/frames/screenshots/webkit-patched-single.png>

Our current understanding of the IE7 policy is that a active frame can navigate a target frame (in the same window) if:

- The target frame is the top-level frame
- The active frame is in the same origin of the target frame or any of its ancestors

So far we have only tried navigating frames within a single window. We're going to work on the multi-window case next.

Comment 7

Mark Rowe (bdrash) — 2007-11-28 21:34:43 PST

Reopening to mark change for review.

Comment 8

Sam Weinig — 2007-11-29 10:43:17 PST

Adam/Collin, Have you tested any special handling of the prenamed targets `_top` and `_parent`?

Comment 9

Adam Barth — 2007-11-29 10:54:46 PST

Have you tested any special handling of the prenamed targets `_top` and `_parent`?

Not yet, but that's a good idea.

We also learned last night that a frame's opener can change after it is created, which complicates the multi-window tests.

Comment 10

Sam Weinig — 2007-11-29 15:56:21 PST

After some testing of the opener frame behavior I have come to the conclusion that a simpler policy may be possible. The policy is:

The navigation change is safe if the active frame is:

- in the same security origin as the target or one of the target's ancestors

Or the target frame is:

- a top-level frame frame in the frame hierarchy

Thoughts?

Comment 11

Sam Weinig — 2007-11-29 17:12:31 PST

Comment on [attachment 17584](https://bugs.webkit.org/attachment.cgi?id=17584&action=edit) [[details]](<https://bugs.webkit.org/attachment.cgi?id=17584&action=edit>) Updates the frame navigation policy to match Internet Explorer 7 (single window)

I have an updated version of this that I will post soon. r- for now

Comment 12

Sam Weinig — 2007-11-29 21:21:32 PST

Additional fix landed in [r28225](#). If you think the policy is still incorrect please let me know.

Comment 13

Adam Barth — 2007-11-30 11:09:40 PST

[r28225](#) looks good to us, and performs the same as IE7 on our test case.

Comment 14

Yuzhu Shen — 2008-01-03 05:06:22 PST

According to the policy implemented in current WebKit, I've created a scenario in which various browsers act differently.

Assume that the following files are located at <http://aaa.bbb.com/> Note: If you want to try out this example, change the value of document.domain accordingly in these files! It should be a suffix of the real domain in which these files reside.

```

=====
<!-- navigation.htm -->
<html>
<head>
  <title></title>
  <script language="javascript">document.domain='bbb.com';</script>
</head>
<frameset name="main" cols="503,*">
  <frame name="left" src="left.htm" />
  <frame name="right" src="right.htm" />
</frameset>
</html>
=====
<!-- left.htm -->
<html>
<head>
  <title></title>
  <base target='right'>
</head>
<body>
  <a href='helloWorld.htm'>HelloWorld</a>
</body>
</html>
=====
<!-- right.htm -->
<html>
<head>
  <title></title>
</head>
<body>
  <script language="javascript">document.domain='bbb.com';</script>
</body>
</html>
=====
<!-- helloWorld.htm -->
<html>
<body>
  <h1>Hello World!</h1>
</body>
</html>
=====

```

If you click the "HelloWorld" link: IE7: will open the link in a new tab instead of the target frame. IE6, Firefox2/3, Safari 3.0.3: will open the link in the target frame. Current WebKit: will do nothing.

This problem affects a popular forum in China: <http://dzh.mop.com> In this forum, if you click a link in the left panel, current WebKit will not open the link. (Due to some other reasons, IE7 works fine with this site.)

Comment 15

David Kilzer (:ddkilzer) — 2008-01-03 09:07:05 PST

(In reply to [comment #14](#))

According to the policy implemented in current WebKit, I've created a scenario in which various browsers act differently.

Please open a new bug for this issue. Thanks!

Comment 16

Yuzhu Shen — 2008-01-03 22:28:05 PST

[bug 16727](#) has been created to track this problem.

Recovery notes

Source evidence recovered on 2026-09-14. The earlier source capture (SHA-256 603e64db34663cbf36f048a0a57f1f9e1b91b08cce0ab21d7653da9fc300d976) is no longer available. This publication uses a separately preserved capture of the same document recorded on 2026-08-17 (SHA-256 26be892a50d53bdac5868c201123a998fb5b69fc545876d39700ef48f5b58110). The article body has been repaired from this source: Initial bug disclosure and complete captured thread restored: report plus comments 1–16. Four-file HTML testcase in comment 14 preserved exactly as fenced source text, including tabs and document.domain assignments. Patch attachment becomes a valid Markdown table with distinct patch-author and reviewer fields. Original reproduction steps, browser-policy comparison, fix revisions and follow-up bug link restored. Report author and source-reported dates now explicit in prepared metadata. The missing earlier capture remains documented in the archive history.

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